

School of Engineering The University of Warwick

General Assignment Information

Module Code	ES97J
Module Title	Biological Systems, Analysis, Dynamics and Control
Assignment Title	Modelling and control of biological processes
Assignment Weighting / Credits	100% / 15 credits Individual Assignment Report 70% / 10.5 credits Individual Assignment Presentation 30% / 4.5 credits
Submission Deadline	Noon, Tuesday, Week 16

Intended Learning Outcomes (ILOs) Assessed

Intended Learning Outcome(s) (ILOs)	Task
Understand fundamental design principles of synthetic biological systems such as modularity, retroactivity, robustness and how modelling can be used in the design of synthetic biological circuits and systems	Part 1
Construct, simulate and analyse complex models of biological systems and understand the role of feedback in regulating biological phenomena	Part 2, 3, 4
Apply an array of computational packages and analysis methods on models and available datasets of diverse biological phenomena.	Part 1
Analyse and design synthetic biological systems by applying MATLAB toolboxes and modelling techniques.	Part 5, 6
Think creatively and beyond traditional discipline boundaries to combine experimental and theoretical concepts, literature and ideas from the field of Synthetic Biology.	Part 7, 8

Notes:

- A total mark below 40% indicates that the ILOs have not all been met at threshold level;
- A total mark in the range 40 – 48% indicates that the ILOs have all been partially met to at least threshold level;
- A total mark of at least 50% indicates that the ILOs have all been met.

Submission Details

The submission details for this assignment are:

- **Deadline:** Noon, Tuesday, Week 16
- **Method:** Report - Online submission via Tabula by deadline above. Presentation slides/material to be emailed to module leader two working days before the given viva date.
- **Format of report:** *A single .pdf file containing the student number, module code and followed by the word assignment, for example: "u123456_ES97J_assignment.pdf"*
- **Submission length:** Maximum 12 pages for report. You will lose **5 marks** per extra page over the limit.
- **Formatting instructions:** Use a minimum 11-point Arial (or equivalent) font for the text, with 1.5 line spacing and 25 mm margins all round. For those who wants to use Latex, ensure that your final pdf outcome is equivalent to the above formatting.

Note: Submissions should be of an appropriate file size and students are responsible for ensuring that work is uploaded successfully before the deadline. If there are technical issues when submitting online, please contact the Engineering Student Office (eng.eso@warwick.ac.uk).

Guidance and Referencing Style

It is serious Academic Misconduct to pass off the work of others (including peers or AI-based chatbots such as ChatGPT) as your own and you should not permit colleagues to copy from you. Sources must be appropriately and properly acknowledged every time reference is made to another's work, using the Harvard Referencing system. Failure to do so amounts to plagiarism which breaches university regulations and falls short of the Academic Integrity expected in the department and university.

Find out more about the School of Engineering Referencing System here:

https://warwick.ac.uk/fac/sci/eng/eso/undergraduate_students/guidance/handbook/skills/s-hb-2-04

There are also other types of academic offences including duplication or 'self-plagiarism'. Refer to <https://warwick.ac.uk/fac/arts/history/students/undergraduate/assess-plagiarism/> for further details.

Style and Formatting Guide

Submissions are expected to conform to professional standards on style and formatting, and guidance can be found here:

https://warwick.ac.uk/fac/sci/eng/eso/undergraduate_students/guidance/handbook/skills/s-h-1-06/

Assignment Feedback

It is generally expected that students will receive the equivalent of 150 – 300 words that covers the strengths of their submission and the areas for improvement. Feedback will be annotated in the submitted pdf file.

ES97J Biological Systems: Analysis, Dynamics and Control

Individual Assignment 2024/2025

Introduction

This research project is concerned with controlling the dynamics of the chemical reaction network shown in Figure 1A. The species X_1 reversibly dimerises to produce X_2 with forward rate β_1 and reverse β_2 . X_2 is converted into X_4 at rate β_3 . X_3 converts into X_5 via two processes; the first is a direct conversion at rate β_5 and the second is via an intermediate X_4 . X_3 is born spontaneously at rate β_0 . Synthetic biologists have engineered the motif in Figure 1B as an “input” into this system in an attempt to get X_5 to track the input μ . This input motif is composed of a single species Z_1 which is produced at rate μ and decays with a reaction rate of η . Z_1 activates the production of X_1 at rate $\alpha \cdot u$, where α is the X_1 production rate and u captures the impact of the regulation of this production by Z_1 . The motif is engineered such that $u = k \cdot Z_1$. As shown in Figure 1C, the system output X_5 tracks the input μ until some disturbance is applied at 150 h. In addition to measuring X_5 , the experimenters have also measured the concentration of X_1 over time. This assignment will design biomolecular controllers to ensure robust tracking.

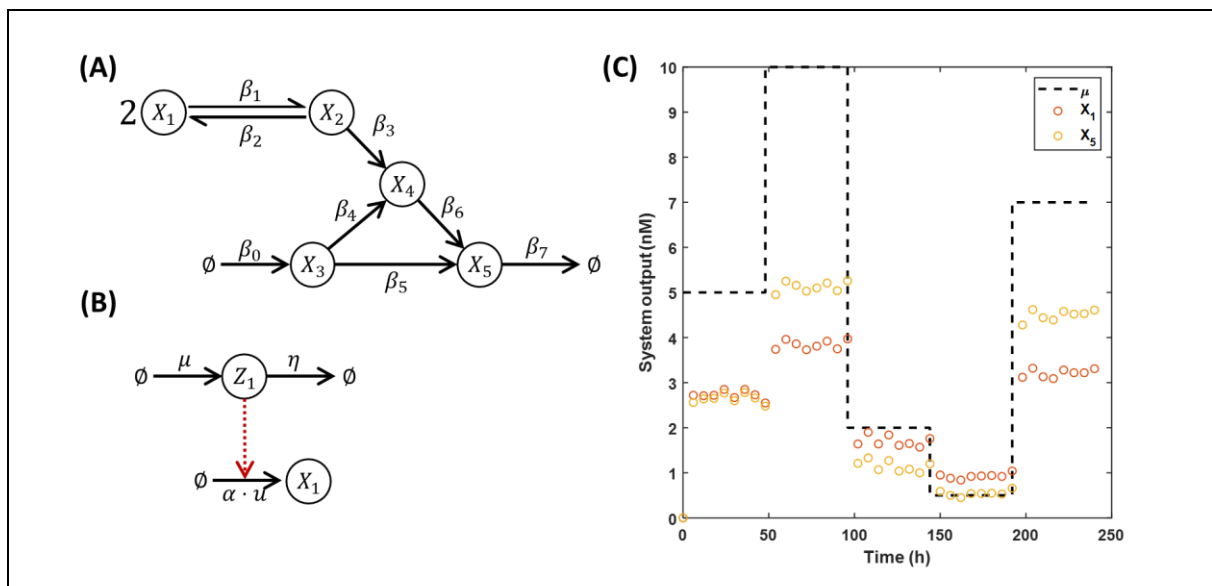


Figure 1. (A) The chemical reaction network. The process network we are seeking to control. **(B) Input motif.** The motif takes input μ and produces chemical species Z_1 . Z_1 activates the birth of species X_1 at rate $\alpha \cdot u$ where $u = k \cdot Z_1$. **(C) The system tracks the input μ until a disturbance is applied.** The input signal is as follows: $0 < t < 48, \mu = 5$; $48 \leq t < 96, \mu = 10$; $96 \leq t < 144, \mu = 2$; $144 \leq t < 192, \mu = 0.5$ and $t \geq 192, \mu = 7$.

Useful Reading

The following references provide some examples of engineering controllers using chemical reaction networks and may prove useful in preparing your assignment.

Aoki, S. K. *et al.* (2019) “A universal rationally-designed biomolecular integral feedback controller for robust perfect adaptation”, *Nature*, 570, pp. 533-537.

Chevalier, M. *et al.* (2019) "Design and analysis of a proportional-integral-derivative controller with biological molecules.", *Cell Systems*, 9(4), ppg. 338-353.

Del Vecchio, D. *et al.* (2016) "Control theory meets synthetic biology." *Journal of the Royal Society Interface*, 13(120), 20160380

Filo, M. *et al.* (2022) "A hierarchy of biomolecular proportional-integral-derivative feedback controllers for robust perfect adaptation and dynamic performance." *Nature Communications*, 13, 2119.

Huang, H. H. *et al.* (2018) "A quasi-integral controller for adaptation of genetic modules to variable ribosome demand." *Nature Communications*, 9, e5415.

Ingalls (2013) "Mathematical Modelling in Systems Biology: An introduction" The MIT Press, Cambridge, MA, USA. Available online at <https://www.math.uwaterloo.ca/~bingalls/MMSB/>.

Qian, Y and Del Vecchio, D. (2018) "Realizing 'integral control' in living cells: how to overcome leaky integration due to dilution?" *Journal of the Royal Society Interface*. 15, 20170902

Section A: Individual Assignment Report

Part 1: Determining the process model (20 marks).

To explore how to control the chemical reaction network, first derive an ordinary differential equation model of the input motif and chemical reaction network shown in Figure 1A and B by applying the law of mass action. The input signal μ is given in the introduction. The rates of the input motif (all in h^{-1}) are: $\alpha = 2$, $\eta = 1$ and $k = 1$. The rates of the chemical reaction network are (also all in h^{-1}) $\beta_0 = 0.5$, $\beta_1 = 1$, $\beta_2 = 2$, $\beta_4 = 2$, $\beta_5 = 1$, $\beta_6 = 3$ and $\beta_7 = 1$. Assume that all species have an initial concentration near zero (such as 10^{-9} nM).

However, the value of β_3 is unknown and needs to be established by data fitting. The values of X_1 and X_5 have been measured and their values up to 144 h are reported in Table 1. Use this information to fit the value of β_3 . You can use the MATLAB function `fminbnd` to do this. You can assume that β_3 has a lower bound of 0 and an upper bound of 20.

Time (h)	0	6	12	18	24	30	36	42	48	54	60	66	72
X_1 (nM)	0.01	2.88	2.73	2.70	2.81	2.84	2.75	2.68	2.60	3.90	3.78	3.69	3.98
X_5 (nM)	0.01	2.72	2.66	2.62	2.74	2.76	2.67	2.61	2.53	5.11	5.07	4.98	5.27
Time (h)	78	84	90	96	102	108	114	120	126	132	138		
X_1 (nM)	3.77	3.85	3.95	3.82	1.84	1.69	1.81	1.82	1.71	1.54	1.66		
X_5 (nM)	5.06	5.14	5.24	5.11	1.41	1.13	1.24	1.25	1.14	0.97	1.10		

Table 1. Experimentally measured values of X_1 and X_5 from Figure 1C.

Once you have correctly calibrated the model, simulate the model's ability to track the input μ . The input signal μ is shown in Figure 1C.

In your report, give details as to how to derive the ordinary differential equation model and give full details of your data fitting approach. Describe, using figures, the internal dynamics of the system.

Part 2: Modelling a disturbance (15 marks).

At $t = 144 \text{ h}$ the system is subjected to a disturbance Δ_1 . This disturbance can be modelled by updating the dynamics of X_2 by replacing the rate β_1 with $(\beta_1 + \Delta_1)$. Leave all other dynamics the same. Simulate the impact of the disturbance $\Delta_1 = 0.5 \text{ h}^{-1}$ at $t = 144 \text{ h}$. This should replicate the observations in Figure 1C.

Vary the size of the disturbance between -0.9 h^{-1} and 1 h^{-1} and discuss the impact of the disturbance on the dynamics of the process's output X_5 , using figures to support your comments.

Add a second disturbance Δ_2 to one of the other rates at a time of your choosing and discuss its impact on the process output X_5 .

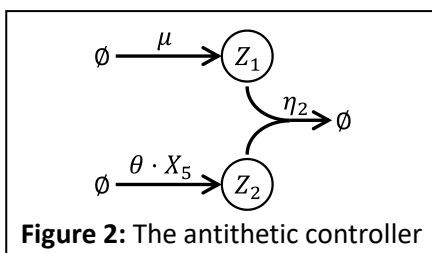
Part 3: Introducing feedback control (10 marks).

We will now design a proportional feedback controller to minimise the impact of Δ . For simplicity we can assume that there is a biological circuit that enables u to be directly calculated as:

$$u = k \cdot Z_1 - K \cdot X_5$$

Discuss how this expression implements proportional negative feedback. Implement this control scheme with $K = 1 \text{ h}^{-1}$ and discuss the impact the controller has on the tracking of μ and its ability to reject both Δ_1 and Δ_2 . Vary K between 0 h^{-1} and 5 h^{-1} and make the design recommendations (paying attention to the size of systems output and the impact of the disturbance).

Part 4: Introducing integral control (10 marks).



Integral control can be realised in living cells using the 'antithetic network' depicted in Figure 2. Implement this control scheme by introducing the new species Z_2 . Note that you will also need to set $\eta = 0 \text{ h}^{-1}$ so that Z_1 only decays through the new reaction in Figure 2 (i.e. via η_2) and not through the reaction in Figure 1B (i.e. not via η)

Initially set $\theta = 1 \text{ h}^{-1}$ and $\eta_2 = 0.1 \text{ h}^{-1}$. Simulate the new controller and discuss how this motif implements integral control.

Vary θ between 0 h^{-1} and 5 h^{-1} and η between 0.01 h^{-1} and 10 h^{-1} and discuss the impact the controller has on the tracking of μ and its ability to reject both Δ_1 and Δ_2 . In your discussion make recommendations as to the choices of θ and η (including making reference the size of systems output, impact of the disturbances and potential impact on the system stability).

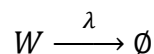
Part 5: PI control for robust tracking (15 marks).

Combine the control strategies from Parts 3 and 4 to implement a PI controller. Using the following parameters $k = 1 \text{ h}^{-1}$, $K = 1 \text{ h}^{-1}$, $\eta = 1 \text{ h}^{-1}$ (for open loop and proportional controllers only), $\eta = 0.1 \text{ h}^{-1}$, $\alpha = 2 \text{ h}^{-1}$, $\theta = 1 \text{ h}^{-1}$ simulate the open loop, proportional, integral and proportional-integral controllers' ability to robustly track the reference signal μ in Figure 1C in light of the disturbances Δ_1 and Δ_2 . Making references to figures, discuss the performance of the three different controllers. You may wish to refer to speed of action, potential overshoots and long-term errors in your discussion.

The analysis has focused on ensuring that X_5 robustly tracks μ . Now copy your model and update the equations so that the biomolecular controller acts on another species X_i , where you choose i . Test the ability of the controllers to make X_i robustly track of μ . Plot the dynamics of all the species in the system and discuss how the different controllers perform. Ensure that you include a disturbance (either Δ_1 , Δ_2 or a new Δ_3) which acts in between the input u and your chosen output species.

Part 6: Incorporating growth (10 marks).

The chemical reaction network shown in Figure 1 neglects to include cell growth. The impact of cell growth at rate λ on a given molecule W can be captured using a simple chemical reaction:



Explain how this additional reaction captures cell growth and update your models to include cell growth at $\lambda = 0.05 \text{ h}^{-1}$.

Re-simulate the performance of the controllers from Part 5 for robust tracking of μ by X_5 . What is the impact of modifying the equations to include growth? Tune k , K , η , η_2 , α and θ as needed to restore good performance. Discuss your observations.

Part 7: Biological realisation (15 marks).

This work has dealt with a theoretical analysis of controllers, which can be implemented using genetic circuits. Several integral controllers have been successfully engineered into living cells by interdisciplinary teams of synthetic biologists and engineers. Conduct a brief literature search to find an example of a synthetic gene circuit which implements integral control that has been designed, built and shown to work in living cells. With the help of a diagram, describe how the circuit you have chosen works in the living cell. Highlight or describe how the disturbance is created and where in the system the integral control occurs. In your description pay attention to how the circuit was realised in the living system. Given synthetic biology is a multi-disciplinary field make sure your description of how integral control is realised and the circuit works is suitable for a mixed audience of biologists and engineers. You should briefly describe what each biological component does, where the biological component originated (e.g., what organism it has been co-opted from) and how it works within the synthetic system.

Part 8: Presentation of Report (5 marks).

It is expected that your report will be presented in a professional manner. Use a minimum 11-point Arial (or equivalent) font for the text, with 1.5 line spacing and 25 mm margins all round. For those who wants to use Latex, ensure that your final pdf outcome is equivalent to the above formatting.

Section B: Individual Assignment Presentation

Section B of this assignment is a 15-minute individual presentation to be delivered over MS Teams at an agreed time in Weeks 17 or 18 (to be confirmed by Module Leader). Using your observations from this assessment and the papers suggested in the bibliography compare the performance of proportional, integral and proportional-integral controllers and how they are impacted by cell growth. Conclude your presentation with an outline of the example you found in Part 6. Please see the final section of this briefing sheet for the marking scheme.

ASSIGNMENT TASKS END.

See overleaf for submission of work and marking scheme.

Submission of work

Submit the following items in Tabula:

- A report (in PDF format, **max 12 pages**) including:
 - A short introduction to the concepts of robust tracking and disturbance and introduction of the chemical reaction network.
 - The derivation of the mathematical model from Part 1.
 - Results –
 - The impact of disturbances on system performance from Part 2
 - Proportional feedback from Part 3
 - Integral feedback from Part 4
 - Performance comparison of the different controllers with conclusions from Part 5
 - Your critical analysis of simulations in Parts 1 to 5 and
 - A discussion of the impact of growth from Part 6.
 - A section on biological realisation of controllers from Part 7.
- MATLAB code needed to simulate the figures in the report as a .zip file. Remember to include all relevant .m files that are required to run the models (including files with ODEs or model parameters).
- Please submit your report as a single pdf file containing the student number, module code and followed by the word assignment, for example: "u123456_ES97J_assignment.pdf"

Marking Scheme

Report (Section A)

Your submitted report will be assessed as follows:

- Part 1. Expected: Equations which describe the dynamics of the system with explanation as to how they were obtained (10 marks). Outline of data fitting approach taken (5 marks). Simulations and discussion of final parameterised model (5 marks).
- Part 2. Expected: Discussion of the impact of the given disturbance on system performance (10 marks) and the student's choice of disturbance (5 marks).
- Part 3. Expected: Discussion of the proportional feedback controller (5 marks) which is supported by appropriate (5 marks).
- Part 4. Expected: Discussion of integral feedback controller (5 marks) supported by appropriate figures (5 marks).
- Part 5. Expected: Discussion of the performance comparison of the controllers with conclusions (10 marks), supported by appropriate figures (5marks).
- Part 6. Expected: Discussion of the impact of growth with references (10 marks).
- Part 7. Expected: A clear description as outlined in Part 7 (10 marks) supported by an appropriate diagram (5 marks).
- Part 8 - Presentation of report (5 marks). Expected: The professionalism of the report that can be understood by others and neatly presented.

The award of marks for each of the parts above follows the university 20 point marking scale, which can be found in the link below.

<https://warwick.ac.uk/services/aro/dar/quality/categories/examinations/markings/ug2017/>

<https://warwick.ac.uk/services/aro/dar/quality/categories/examinations/markings/ug2017/>

Presentation (Section B)

DELIVERY OF SUBJECT CONTENT (50%)	DESIGN AND ENGAGEMENT WITH VISUAL AID (25%)	ANSWERS TO QUESTIONS (15%)	ORAL DELIVERY (10%)
70-100. Outstanding subject knowledge. Extremely comprehensive and thorough presentation of the content. Selection of key points is outstanding with minimal or no omissions. All obvious topics and angles are covered. Ownership of central arguments is made clear. Attribution to all external sources is appropriately and consistently provided (in-text and reference list).	70-100. Exemplary slide design. Extremely visually attractive, engaging and accessible. Well-structured slides, fully supporting delivery with key points. Excellent balance of text and other media. Points on slides exceptionally well-integrated into the delivery, in which new, relevant information is appropriately tailored to audience needs. Demonstrated comprehensive understanding of content and seemed fully prepared.	70-100. Able to articulate thorough, comprehensive responses that demonstrate an exceptionally high and well-rounded understanding of the topic. Responses include detailed and highly relevant examples, sources and/or facts to support arguments. Answers demonstrate a strong critical awareness of key issues surrounding the topic, including (where relevant) a heightened ability to critically self-reflect.	70-100. Very engaging and exemplary delivery style. Able to communicate all ideas exceptionally well to a wide and diverse audience. Very clear and appropriate ion and tone of voice. Excellent pacing throughout, including a balanced delivery time between slides. Explained all terms, acronyms & jargon appropriately. Fluid delivery with no unnecessary pauses. Exceptional timing (9 minutes +/- 30 seconds).
60-69. Very good subject knowledge demonstrated. Detailed presentation of the content. Selection of key points is good, though with some minor omissions. Most topics and angles are covered well. There is some ownership in the central arguments that are made, though with some lack of clarity. Attribution to external sources is provided in most cases, with consistency, however with some minor errors (in-text and reference list).	60-69. High standard of slide design. Relatively visually attractive, engaging and accessible, with minor errors (e.g. font too small). Slides are generally well-structured and support the key points and delivery relatively well. Use of text and other media is generally appropriate, though may have missed an opportunity to use a photo/diagram in the place of text. At times, audience needs are not considered. Demonstrated a good understanding of content and generally seemed prepared.	60-69. Able to articulate detailed and full responses that demonstrate a high and generally well-rounded understanding of the topic. Some responses include relevant examples, sources and/or facts to support arguments. Most answers demonstrate a critical awareness of key issues surrounding the topic, including (where relevant) an ability to self-reflect on progress.	60-69. Engaging delivery style presented to high standards. Able to communicate most ideas well to a wide and diverse audience. Clear and generally appropriate ion and tone of voice. Good pacing throughout, including a relatively balanced delivery time between slides. Explained most terms, acronyms & jargon appropriately. Generally fluid delivery with minimal unnecessary pausing. Generally good timing (8 minutes +/- 1 minute).
50-59. Sufficient subject knowledge demonstrated. content is adequately presented. Selection of key points is sufficient, though some significant elements have been omitted. Topics and angles have satisfactorily been covered, though lacking depth or breadth. Central	50-59. Slide design meets threshold requirements, however the design lacks clarity, coherence and/or includes some omissions. Some design weaknesses (e.g. too much or too little content on the slides). Slides are only adequately structured and sometimes do not	50-59. Able to articulate responses that demonstrate a basic understanding of the topic. Some responses may include relevant examples, sources and /or facts to support arguments, though not in much depth. Answers adequately demonstrate a critical awareness of key	50-59. Satisfactory delivery style which presented some areas of weakness. Able to communicate ideas adequately, though the diversity of audience members may not have been considered at some points. Sufficient ion and tone of voice. Pacing is satisfactory, though at

DELIVERY OF SUBJECT CONTENT (50%)	DESIGN AND ENGAGEMENT WITH VISUAL AID (25%)	ANSWERS TO QUESTIONS (15%)	ORAL DELIVERY (10%)
arguments of the are minimally illuminated. Attribution to external sources is limited and/or lacking in consistency in some areas. Some errors (in-text and reference list).	support the key points and delivery. Balance of text and other media is satisfactory. Little consideration of audience needs. Demonstrated a weak understanding of content, and there is evidence of a lack of preparation.	issues surrounding the topic, including (where relevant) a minimal degree of self-reflection on progress.	times presented an unbalanced delivery time between slides. Some terms, acronyms & jargon are explained, though may be inconsistent or lack clarity. Some hesitation and unnecessary pausing. Timing is satisfactory (6 minutes +/- 1 minute).
40-49. Superficial level of subject knowledge demonstrated. content is poorly presented. Selection of key points is weak, leading to many omissions. Topics and angles are not sufficiently covered in any depth or breadth. Central arguments are unclear or poorly articulated. Attribution to external sources is poor, though nonetheless made, highlighting some significant academic practice issues. Significant errors and/or some omissions (in-text and reference list).	40-49. Slide design significantly lacks clarity and coherence, and has several missing elements. Slides are inadequately structured, which has impacted its delivery. Integration between slide and oral delivery is unclear. Use of text and other media in the slides contains many errors or is somewhat irrelevant. Minimal consideration of audience needs. Demonstrated a limited understanding of the content, and the lack of preparation distracts from the content being delivered.	40-49. Limited ability to articulate responses that demonstrate an understanding of the topic. Most responses do not include any relevant examples, sources and/or facts to support points or arguments. There is an limited level of critical awareness on key issues surrounding the topic. Limited degree of self-reflection on progress.	40-49. Inappropriate delivery style presenting a number of weaknesses. Communication of ideas is strained and consideration of audience diversity is limited. Pacing is often inadequate, with a noticeable imbalance of delivery time between slides. Some attempt has been made to explain some terms, acronyms and &/or jargon, though limited. Frequent hesitation and unnecessary pausing. Timing is insufficient to meet threshold (5 minutes or > 10:30 +/- 1 minute).
30-39. Insufficient level of subject knowledge. Significant gaps in presentation of content. Selection of key points is poor and seemingly sporadic, leading to a substantial amount of omissions. Limited (if any) discussion of topics and angles in any depth or breadth. Minimal or no clear arguments presented. Very poor or no attribution to any external sources made. Substantial academic practice issues and/or omissions present (in-text and reference list). Evidence of plagiarism.	30-39. Slide design is confusing, with minimal evidence of any coherence. Large sections are missing. Slides show little to no structure. There is minimal (if any) interaction between slides and the oral delivery, or slides are missing entirely. Limited or no use of text and/or other media in the slides, or visuals are entirely irrelevant to the content. No consideration of audience needs. Demonstrated minimal to no understanding of the content, and seems entirely unprepared.	30-39. Unable to demonstrate an understanding of the topic. Limited to no references made to examples, sources and/or facts. No argumentation or critical awareness on the topic. Unable to self-reflect on progress.	30-39. Delivery style entirely inappropriate. Minimal communication of ideas, and audience diversity is not considered. Minimal attention to pacing and demonstrates an unfamiliarity with the content. No attempt made to explain terms, acronyms and/or jargon. Many hesitations and unnecessary pausing. Timing is extremely poor (< 3 minutes).

